

with the overall very weak predictability in data (see chapter 8), any realistic application of Figure 4.8 considerably flattens both the time-varying short-run and opportunistic weights as a function of expected returns.

3. Rebalancing is Short Volatility

Rebalancing is an option strategy and, in particular, a *short volatility strategy*. (Option traders would call rebalancing a *negative gamma* strategy.) This is not well known, although at some level should not be surprising for the reader steeped in financial theory because the same method used in section 2 to solve long-horizon portfolio choice problems (dynamic programming) is used to value options (where it is called *backward induction*).²⁶ Showing how rebalancing is mechanically a short volatility strategy gives us deeper insights into what long-run investors are gaining and losing from rebalancing. Nothing is free after all, at least not in economic theory.

3.1. EXAMPLE

This example is highly stylized and simple but conveys enough to see rebalancing as a collection of options.

Suppose that a stock follows the *binomial tree* given in Figure 4.9, Panel A. Each period the stock can double, with probability 0.5, or halve starting from an initial value of $S = 1$. There are two periods, so there are three nodes in the tree. At maturity, there are three potential payoffs of the stock: $S_{uu} = 4$, $S_{ud} = S_{du} = 1$, and $S_{dd} = 0.25$, which have probabilities of 0.25, 0.5, and 0.25, respectively. In addition, the investor can hold a risk-free bond that pays 10% each period.

Let us first consider a buy-and-hold strategy that starts out with 60% equities and 40% in the risk-free asset. (We know buy and hold is not optimal for the long-run investor from section 2.) At the end of the first period, the wealth of this investor can increase or decrease to

$$\begin{aligned} W_u &= 0.6 \times 2.0 + 0.4 \times 1.1 = 1.6400 \\ \text{or } W_d &= 0.6 \times 0.5 + 0.4 \times 1.1 = 0.7400, \end{aligned} \tag{4.18}$$

which is shown by branching of the binomial tree in Figure 4.9, Panel B. In equation (4.18), the return on the stock is either $2 - 1 = 100\%$ if we go into the upper branch or $0.5 - 1 = -50\%$ if we go into the lower branch. In the upper node at time 1, the proportion of the buy-and-hold portfolio held in equities is $0.6 \times 2.0 / 1.64 = 73.17\%$ and the proportion of the portfolio in equities in the

²⁶ Perold and Sharpe (1988) and Cochrane (2007) discuss interpreting rebalancing as an option strategy.